Haesung Kim

Ph.D. in Electrical Engineering

Postdoctoral Researcher
Specere Lab, School of Mechanical Engineering, Birck Nanotechnology Center, Purdue University, West Lafayette, IN

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Research Focus

The work develops characterization techniques and physics-based models that separate device parameters which conventional measurements leave entangled. Methods established in this line extract parasitic source and drain resistances independently of the threshold voltage, the mobility enhancement factor, interface-trap and subgap density-of-states distributions, and the conduction band minimum as a robust energy reference for amorphous oxide semiconductors. The approach rests on opto-electronic measurement — photonic I–V and C–V with the photovoltaic and photoconductive contributions de-embedded — together with transient and pulsed characterization. Recent work applies these methods to emerging memory devices: charge-trapping dynamics and read-after-write latency in $HfZrO_x$ ferroelectric FETs, trap dynamics resolved from the transient response of a-IGZO thin-film transistors, low-frequency noise as a function of contact metal, intrinsic field-effect mobility at cryogenic temperature, and proton-irradiation damage at SiN_x and Si–SiO_2 interfaces in flash memory. Current work at Purdue extends this line to ferroelectric $HfZrO_x$: how the crystallization anneal route shapes the ferroelectric response, the mechanisms behind endurance degradation and how that degradation varies spatially across a device population, photo-toggled capacitance-transient spectroscopy that resolves the subgap density of states in both space and energy, and the reliability of two-dimensional transistors under bias stress in an industrial collaboration.

Education

2022.03 – 2025.08	**Ph.D. in Electrical Engineering** Kookmin University, Seoul, Korea Advisors: Prof. Dong Myong Kim, Prof. Jong-Ho Bae (research) Prof. Sung-Jin Choi (advisor of record)
2020.03 – 2022.02	**M.S. in Electrical Engineering** Kookmin University, Seoul, Korea
2014.03 – 2020.02	**B.S. in Electrical Engineering** Kookmin University, Seoul, Korea

Appointments

2025.08 – present	**Postdoctoral Researcher** Specere Lab, School of Mechanical Engineering Birck Nanotechnology Center, Purdue University, West Lafayette, IN, USA Advisor: Prof. Thomas Beechem
2025.03 – 2025.06	**Lecturer** Kookmin University, Seoul, Korea Intelligent Semiconductor Devices (3 credits, junior level)

Military Service

2015.09 – 2017.06	**Republic of Korea Marine Corps** — mandatory military service

International Journal Articles

1. Y. Choi, H. Yang, H. J. Kim, S. Park, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, **H. Kim**, J. H. Bae, "Characterization of trap dynamics via transient response in amorphous InGaZnO thin-film transistors," *Applied Physics Letters*, vol. 128, no. 24, 2026. doi:10.1063/5.0332203 **(Corresponding Author)**

2. H. J. Kim, H. Yang, **H. Kim**, S. Park, H. N. Lee, S. J. Choi, D. M. Kim, D. H. Kim, M. K. Park, J. H. Bae, "Analysis of spatial distribution of remnant polarization in inverted staggered a-IGZO/HZO ferroelectric thin-film transistor using capacitance-voltage characteristics," *Solid-State Electronics*, vol. 235, pp. 109358, 2026. doi:10.1016/j.sse.2026.109358
3. **H. Kim**, J. Y. Park, S. H. Han, H. Yang, Y. J. Lee, S. J. Choi, D. H. Kim, J. H. Bae, D. M. Kim, "Characterization of Low-Temperature Intrinsic Field-Effect Mobility With Contact Resistances in Amorphous InGaZnO Thin Film Transistors," *IEEE Transactions on Electron Devices*, vol. 73, no. 7, pp. 4341–4345, 2026. doi:10.1109/TED.2026.3691120 **(First Author)**
4. **H. Kim**, J. Park, J. Im, H. Kim, Y. M. Kim, K. Park, S. J. Choi, D. H. Kim, D. M. Kim, Y. J. Yoon, S. Y. Woo, J. H. Bae, "Impact of proton irradiation on SiN_x and Si–SiO_2 interfaces in FLASH memory cell," *Nuclear Engineering and Technology*, vol. 58, no. 10, pp. 104433, 2026. doi:10.1016/j.net.2026.104433 **(First Author)**
5. Y. Kim, Y. J. Lee, J. Yang, B. Kim, S. J. Kim, J. Kim, I. Im, J. Y. Kim, H. Rui, **H. Kim**, C. W. Bark, M. H. Park, D. H. Kim, S. J. Choi, J. J. Yang, S. Lee, H. W. Jang, "Programmable ferroelectric rectifier for reliable and efficient neuromorphic crossbar array," *Nature Communications*, vol. 17, no. 1, 2026. doi:10.1038/s41467-026-70727-2
6. S. Kim, H. Yang, H. Lee, J. Park, H. Jeong, S. Y. Woo, M. K. Park, **H. Kim**, J. H. Bae, "Source/drain metal work-function-dependent memory characteristics governed by ferroelectric switching-assisted current modulation in a-IGZO FeFETs," *Materials Science in Semiconductor Processing*, vol. 215, pp. 111021, 2026. doi:10.1016/j.mssp.2026.111021
7. **H. Kim**, H. Yang, Y. Choi, H. Kim, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Unveiling charge trapping dynamics in SiO_2/$HfZrO_X$–based ferroelectric field–effect–transistors to minimize read–after–write latency," *Current Applied Physics*, vol. 89, pp. 181–185, 2026. doi:10.1016/j.cap.2026.06.007 **(First Author)**
8. H. Lee, S. Kim, C. Han, **H. Kim**, H. Yang, S. Park, S. Yun, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Quantitative Analysis on the Interaction Between Channel Carrier and Remote Trap in $Hf_xZr_{1-x}O_2$/SiO_2 Interface in Ferroelectric Field-Effect-Transistor," *Advanced Electronic Materials*, vol. 12, no. 2, 2026. doi:10.1002/aelm.202500549
9. J. Park, **H. Kim**, S. J. Choi, D. H. Kim, D. M. Kim, J. H. Bae, "Exploring low-frequency noise dynamics in a-IGZO TFTs: Unveiling the impact of contact metal variations for advanced semiconductor applications," *Solid-State Electronics*, vol. 231, pp. 109271, 2026. doi:10.1016/j.sse.2025.109271 **(Co-first Author)**
10. H. Yang, **H. Kim**, H. J. Kim, D. M. Kim, S. J. Choi, D. H. Kim, Y. J. Lee, J. H. Bae, "Defect-mediated enhancement of memory window in IGZO-channel ferroelectric field effect transistors," *Materials Science in Semiconductor Processing*, vol. 204, pp. 110268, 2026. doi:10.1016/j.mssp.2025.110268
11. **H. Kim**, S. H. Han, H. Jeong, H. Yang, S. J. Choi, D. H. Kim, J. H. Bae, D. M. Kim, "Photonic I–V Technique With Photogating Effect for Extended Extraction of Subgap Density of States in Amorphous Oxide Semiconductor TFTs," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2751–2755, 2025. doi:10.1109/TED.2025.3547707 **(First Author)**
12. S. Park, H. Yang, **H. Kim**, H. Jeong, S. J. Choi, D. H. Kim, D. M. Kim, M. K. Park, J. H. Bae, "Analysis of operation delay and switching speed limitation due to source/drain resistance and subgap density of states in amorphous $InGaZnO_x$/$HfZrO_x$ ferroelectric thin-film transistor," *Solid-State Electronics*, vol. 229, pp. 109203, 2025. doi:10.1016/j.sse.2025.109203
13. S. H. Han, **H. Kim**, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Photovoltaic Effect De-Embedded Photonic C–V Characterization of Subgap Density of States in Amorphous Oxide Semiconductor Thin-Film Transistors," *IEEE Transactions on Electron Devices*, vol. 71, no. 11, pp. 6795–6798, 2024. doi:10.1109/TED.2024.3469161 **(Co-first Author)**
14. **H. Kim**, H. Yang, S. Lee, S. Yun, J. Park, S. Park, H. N. Lee, H. Kim, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Comprehensive analysis of read-after-write latency in $HfZrO_X$-based ferroelectric field-effect-transistors with SiO_2 interfacial layer," *Applied Physics Letters*, vol. 124, no. 3, 2024. doi:10.1063/5.0172204 **(First Author)**
15. S. Lee, **H. Kim**, H. Yang, S. Yun, J. Park, H. Lee, S. Park, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Analysis of the Role of Interfacial Layer in Ferroelectric FET Failure as a Memory Cell," *IEEE Electron Device Letters*, vol. 45, no. 4, pp. 562–565, 2024. doi:10.1109/LED.2024.3360419 **(Co-first Author)**
16. H. Yang, S. Park, S. Yun, **H. Kim**, H. Lee, M. K. Park, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Exploration of structural influences on the ferroelectric switching characteristics of ferroelectric thin-film transistors," *Nanoscale*, vol. 16, no. 42, pp. 19856–19864, 2024. doi:10.1039/d4nr02096k
17. **H. Kim**, H. B. Yoo, H. Lee, J. H. Ryu, J. Y. Park, S. H. Han, H. Yang, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Extraction Technique for the Conduction Band Minimum Energy in Amorphous Indium–Gallium–Zinc–Oxide Thin

Film Transistors," *IEEE Transactions on Electron Devices*, vol. 70, no. 6, pp. 3126–3130, 2023. doi:10.1109/TED.2023.3269735 **(First Author, Equal Contribution)**

18. J. Y. Park, **H. Kim**, H. B. Yoo, J. H. Ryu, S. H. Han, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Simultaneous Extraction of Mobility Enhancement Factor and Threshold Voltage With Parasitic Resistances in Amorphous Oxide Semiconductor Thin-Film Transistors," *IEEE Transactions on Electron Devices*, vol. 70, no. 5, pp. 2312–2316, 2023. doi:10.1109/TED.2023.3253468 **(Co-first Author)**

19. **H. Kim**, H. B. Yoo, J. H. Ryu, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Current-to-transconductance ratio technique for simultaneous extraction of threshold voltage and parasitic resistances in MOSFETs," *Solid-State Electronics*, vol. 183, pp. 108133, 2021. doi:10.1016/j.sse.2021.108133 **(First Author, Equal Contribution)**

20. H. B. Yoo, **H. Kim**, J. H. Ryu, J. Park, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Modeling and characterization of photovoltaic and photoconductive effects in insulated-gate field effect transistors under optical excitation," *Solid-State Electronics*, vol. 186, pp. 108139, 2021. doi:10.1016/j.sse.2021.108139

21. J. Yu, H. B. Yoo, **H. S. Kim**, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, "Characterization of Spatial Distribution of Trap Across the Substrate in Metal-Insulator-Semiconductor Structure with Band Bending Effect," *Journal of Nanoscience and Nanotechnology*, vol. 21, no. 8, pp. 4315–4319, 2021. doi:10.1166/jnn.2021.19391

22. **H. Kim**, H. B. Yoo, J. T. Yu, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, "Alternating Current-Based Technique for Separate Extraction of Parasitic Resistances in MISFETs With or Without the Body Contact," *IEEE Electron Device Letters*, vol. 41, no. 10, pp. 1528–1531, 2020. doi:10.1109/LED.2020.3020405 **(First Author, Equal Contribution)**

23. H. B. Yoo, J. Yu, **H. Kim**, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, "Deep depletion capacitance–voltage technique for spatial distribution of traps across the substrate in MOS structures," *Solid-State Electronics*, vol. 173, pp. 107905, 2020. doi:10.1016/j.sse.2020.107905

International Conference Papers

1. H. Yang, **H. Kim**, C. Han, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Quantitative Analysis of Endurance-Dependent Charge Trapping Dynamics in Ferroelectric Field-Effect Transistors with Temperature Effects," *2025 Silicon Nanoelectronics Workshop (SNW)*, pp. 74–75, 2025. doi:10.23919/SNW65111.2025.11097276

2. H. B. Yoo, **H. Kim**, Y. Lee, J. H. Ryu, J. Y. Park, H. J. Yang, J. H. Bae, D. H. Kim, S. J. Choi, D. M. Kim, "Characterization Technique for Interface Traps in Si Nanosheet GAA MOSFETs through Subthreshold I-V Characteristics," *2022 IEEE 22nd International Conference on Nanotechnology (NANO)*, pp. 116–119, 2022. doi:10.1109/NANO54668.2022.9928778 **(Co-first Author)**

3. **H. Kim**, S. H. Han, H. Jeong, H. Yang, S.-J. Choi, D. H. Kim, J.-H. Bae, D. M. Kim, "Photonic I–V-based technique for subgap density of states extraction in AOS TFTs," *International Conference on Electronics, Information, and Communication (ICEIC)*, Osaka, Japan, Jan. 2025.

4. **H. Kim**, J. Park, J. Im, H. Kim, Y. J. Yoon, Y.-M. Kim, K. Park, S. Y. Woo, J.-H. Bae, "Impact of Proton Irradiation on SiN_x and Si–SiO_2 Interfaces in FLASH Memory," *IEEE Silicon Nanoelectronics Workshop (SNW)*, Honolulu, HI, USA 2024, pp. 1–2.

Domestic Conference Papers (Korea)

1. J. Park, S. Lee, **H. Kim**, H. Jeong, Y. Choi, S.-J. Choi, D. M. Kim, D. H. Kim, J.-H. Bae, "Comparative Analysis of the Low-Frequency Noise Behavior of a-IGZO TFT with Different Source/Drain Metal," *The 31st Korean Conference on Semiconductors*, Jan. 2024.

2. Y. Choi, **H. Kim**, H. Yang, S. Park, J. Park, S.-J. Choi, D. H. Kim, D. M. Kim, J.-H. Bae, "Analysis on Drain Current Transient Response in Amorphous $InGaZnO_x$ Thin-Film Transistors," *The 31st Korean Conference on Semiconductors*, Jan. 2024.

3. S. H. Han, **H. Kim**, J. Y. Park, J.-H. Bae, S.-J. Choi, D. H. Kim, D. M. Kim, "Extraction of Subgap Density-of-States in AOS TFTs through Capacitance-Voltage Characteristics Considering Photovoltaic Effect," *The 31st Korean Conference on Semiconductors*, Jan. 2024.

4. H.-N. Lee, H. Yang, S. Park, **H. Kim**, S. Yun, S.-J. Choi, D. H. Kim, D. M. Kim, J.-H. Bae, "Analysis and Modeling of Ferroelectric Amorphous $InGaZnO_x$ Thin-Film Transistor at Initial State and During Memory Operation," *The 31st Korean Conference on Semiconductors*, Jan. 2024.
5. S. Park, H. Yang, **H. Kim**, S. Yun, H.-N. Lee, D. M. Kim, D. H. Kim, S.-J. Choi, J.-H. Bae, "Comparative Analysis of Polarization Switching Characteristics in Channel and Contact Regions of Ferroelectric $InGaZnO_x$ Thin Film Transistor," *The 31st Korean Conference on Semiconductors*, Jan. 2024.
6. S. J. Kim, H.-N. Lee, H. J. Yang, **H. Kim**, S. J. Park, S.-J. Choi, D. M. Kim, D. H. Kim, J.-H. Bae, "Analysis of interface state according to the polarization switching of ferroelectric field-effect transistor," *The 31st Korean Conference on Semiconductors*, Jan. 2024.
7. H. B. Yoo, **H. Kim**, J. Ryu, J. Y. Park, S. H. Han, H.-I. Yang, J.-H. Bae, S.-J. Choi, D. H. Kim, D. M. Kim, "Characterization of Lateral Trap Distribution in a-InGaZnO TFTs through Capacitance-Voltage Technique Combined with Extended Channel Conduction Factor," *The 30th Korean Conference on Semiconductors*, Feb. 2023.
8. J. Park, S. Lee, **H. Kim**, S.-J. Choi, D. M. Kim, D. H. Kim, J.-H. Bae, "Optimization of Memory Properties of 2T-DRAM Cell by a-IGZO TFT Engineering," *The 30th Korean Conference on Semiconductors*, Feb. 2023.
9. J. Y. Park, H. B. Yoo, **H. Kim**, J. H. Ryu, S. H. Han, J.-H. Bae, S. Choi, D. H. Kim, D. M. Kim, "Extraction Technique for Characteristic Parameters in Si MOSFETs through the Dual Sweep Current-to-Transconductance Ratio," *The 30th Korean Conference on Semiconductors*, Feb. 2023.
10. S. Park, H. Yang, **H. Kim**, S. Yun, D. M. Kim, D. H. Kim, S.-J. Choi, J.-H. Bae, "Effect of IGZO/HfO_2 Ferroelectric Capacitor Structure on Polarization Switching," *The 30th Korean Conference on Semiconductors*, Feb. 2023.
11. S. H. Han, H. B. Yoo, **H. Kim**, J. Ryu, J. Y. Park, J.-H. Bae, S.-J. Choi, D. H. Kim, D. M. Kim, "Characterization of Photovoltaic and Photoconductive Responses in Amorphous Oxide Semiconductor Thin-Film Transistors," *The 30th Korean Conference on Semiconductors*, Feb. 2023.
12. S. Lee, **H. Kim**, H. Yang, S. Yun, S.-J. Choi, D. H. Kim, D. M. Kim, J.-H. Bae, "Analysis of Memory Window Degradation Mechanism by Unipolar and Bipolar Stress in HfO_2-based FeFETs," *Summer Annual Conference of the Institute of Electronics and Information Engineers*, Jun. 2022.
13. S.-H. Yun, **H. Kim**, H. Yang, S. Lee, S.-J. Choi, D. H. Kim, D. M. Kim, J.-H. Bae, "Analysis on Leakage Current of Gated Diode Memory Device and Suggestion of Improved Device Structure," *Summer Annual Conference of the Institute of Electronics and Information Engineers*, Jun. 2022.
14. H. Yang, **H. Kim**, S. Yun, S. Lee, D. M. Kim, D. H. Kim, S.-J. Choi, J.-H. Bae, "Effect of Annealing Sequence on the a-IGZO FeTFT Ferroelectric Switching Characteristics," *Summer Annual Conference of the Institute of Electronics and Information Engineers*, Jun. 2022.
15. J. Y. Park, **H. Kim**, H. B. Yoo, J. H. Ryu, J.-H. Bae, S.-J. Choi, D. H. Kim, D. M. Kim, "Comprehensive DC Technique for Extraction of Threshold Voltage, Parasitic Resistances, and Mobility Enhancement Parameter in AOS TFTs," *The 28th Korean Conference on Semiconductors*, Jan. 2021. **(Co-first Author)**
16. J. H. Ryu, H. B. Yoo, J. Yu, **H. Kim**, J.-H. Bae, S.-J. Choi, D. H. Kim, D. M. Kim, "Extraction of Interface Traps over the Bandgap through Photovoltaic and Photoconductive Effects in Si MOSFETs under Optical Excitation," *The 28th Korean Conference on Semiconductors*, Jan. 2021.
17. H. B. Yoo, **H. Kim**, J. Yu, Y. J. Park, S.-J. Choi, D. H. Kim, D. M. Kim, "Alternating Current-based Open Drain Method for Separate Extraction of Source and Drain Resistances in MOSFETs," *The 27th Korean Conference on Semiconductors*, Feb. 2020. **(Co-first Author, Presenter)**
18. J. Yu, H. B. Yoo, **H. Kim**, Y. J. Park, S.-J. Choi, D. H. Kim, D. M. Kim, "Characterization of Spatial Distribution of Traps across the Substrate in Metal-Insulator-Semiconductor Structures with Band Bending Effect," *The 27th Korean Conference on Semiconductors*, Feb. 2020.
19. H. B. Yoo, Y. J. Park, J. Yu, **H. Kim**, S.-J. Choi, D. H. Kim, D. M. Kim, "Modeling and Characterization of the Photovoltaic and Photoconductive Effects in Field Effect Transistors under Optical Illumination," *The 27th Korean Conference on Semiconductors*, Feb. 2020.

Preprints

1. D. M. Kim, J. H. Ryu, H. B. Yoo, **H. Kim**, J. Y. Park, S. H. Han, J. H. Bae, S. J. Choi, D. H. Kim, "Capacitance-Based Extraction Technique for the Spatial and Energy Distributions of Traps in Si Mosfets," *SSRN*, 2023. doi:10.2139/ssrn.4377528

Patents

2026	**Method and apparatus for extracting the conduction band minimum of amorphous oxide semiconductor thin-film transistors** (translated; official title in Korean) KR Patent **10-2914693** · registered 2026-01-14 · filed 2023-03-17 (10-2023-0035106) Assignee: Kookmin University Industry-Academic Cooperation Foundation · 11 claims · inventor 5 of 10

Research Grants and Projects

2024-09 – 2025-08	**Principal Investigator** — National Research Foundation of Korea Maximizing Energy Efficiency and Speed: Modeling and Optimization of Transient Response Characteristics in AOS-based Ferroelectric Memory Devices
2024-03 – 2024-12	**Graduate Researcher** — National Research Foundation of Korea Development of Application-customized Computing Be-spoke Device for High-density and Efficient PIM
2022-06 – 2023-05	**Graduate Researcher** — Industry–University Cooperation Research, Alsemy Co. Semiconductor device modeling using artificial neural networks
2020 – 2024	**Graduate Researcher and Staff** — National Research Foundation of Korea Four-Step Brain Korea 21 Project
2020 – 2022	**Graduate Researcher** — National Research Foundation of Korea Hybrid Device-Based Circadian ICT Research Center

Awards and Honors

2025	**Excellent Talent Award**, Kookmin University (Ph.D. conferral, Fall 2025)
2023	**Excellent Poster Award on Site**, The 30th Korean Conference on Semiconductors
2022	**Excellent Talent Award**, Kookmin University (M.S. conferral, Spring 2022)

Teaching Experience

2025	**Lecturer** — Intelligent Semiconductor Devices, Kookmin University (3 credits, junior level; chalkboard lecture. Reference: Neamen, Semiconductor Physics and Devices)
2024	**Session Instructor** — Winter job training: Measurement of semiconductor devices and extraction of characteristic parameters, Kookmin University (senior level)
2023	**Session Instructor** — Summer job training: Measurement of semiconductor devices and extraction of characteristic parameters, Kookmin University (senior level)
2023	**Teaching Assistant** — Basic Electronic Experiment, Kookmin University (sophomore level)
2023	**Teaching Assistant** — Electrical Engineering Capstone Design I, Kookmin University (senior level)
2022	**Teaching Assistant** — Electrical Engineering Capstone Design II, Kookmin University (senior level)
2022	**Teaching Assistant** — Electrical Engineering Capstone Design I, Kookmin University (senior level)
2021	**Teaching Assistant** — Electrical Engineering Capstone Design II, Kookmin University (senior level)
2021	**Session Instructor** — Summer job training: Semiconductor Device TCAD Simulation Training, Kookmin University (junior–senior level)
2020	**Teaching Assistant** — Electrical Engineering Capstone Design II, Kookmin University (senior level)

2019	**Undergraduate Mentoring** — Semiconductor Engineering, Kookmin University (junior level. Reference: D. M. Kim, Microelectronic Semiconductor Physics and Devices)
2019	**Undergraduate Mentoring** — Electromagnetics, Kookmin University (sophomore level. Reference: W. Hayt, Engineering Electromagnetics)

Equipment Experience

Agilent 4156C	Semiconductor parameter analyzer
Agilent B1500A	Semiconductor device analyzer
Keithley 4200-SCS	Semiconductor parameter analyzer
HP 4284A	LCR meter (20 Hz–1 MHz)
HP 4294A	LCR meter (40 Hz–100 MHz)
Keysight 35670A	Dynamic signal analyzer
SR570	Low-noise current preamplifier
CX3322A	Device current waveform analyzer
RIGOL DG5102	100 MHz 2-channel arbitrary function generator
DPO3014	4-channel 100 MHz digital oscilloscope
Agilent E5250A	Switching matrix
SUMMIT-9200	Probe station
MS TECH MST-1000B	Hot chuck controller
OZ-Series	Laser diode sources
PM100D	Digital optical power and energy meter console
WITec alpha	Raman and PL spectroscopy

Software and Simulation Tools

Synopsys ISE TCAD	Device simulation
Silvaco TCAD	Process and device simulation
OrCAD PSpice Designer	SPICE simulator for analog and mixed-signal design and verification
PSIM	Power electronics simulation
P–CAD	EDA software for PCB design
MATLAB	Data processing and analysis
Python	Data analysis and automation scripting

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